

Temporal Semantic Discovery for Progressive Retrieval Attention

Look-Behind, Delayed Commitment, and Look-Ahead Routing

Jorge Simão

EInnovator R&D – Center for the Sciences of Computation, Cognition, and Complexity

August 2026

Abstract

Progressive Retrieval Attention (PRA) separates logical-memory discovery from native-K/V materialization, but semantic discovery commonly treats one current token or one compressed query state as the routing unit. Language is not naturally organized at that granularity: adjacent tokens are correlated, names and relations often become discriminative only after several tokens, and memory chunks may themselves be represented by only one or a few gists. We study temporal semantic discovery: routing from short spans of past, current, and, when available, future token representations. We separate causal look-behind with delayed commitment, prompt-prefill look-ahead through a non-causal routing sidecar, and optional predictive look-ahead during generation. Paper 2.6 indexed lexical retrieval remains the preferred path for explicitly structured memories such as tool/API registries. The central hypothesis is that retrieval can operate on a slower span-level clock while autoregressive generation remains token-causal.

1 Position in the PRA series

Paper 2.5 establishes iterative associative traversal and separates root entry, propagation, stopping, and query formulation. Paper 2.6 establishes multi-representational discovery, including exact-token, n-gram, BM25, approximate, sparse indexed, and hybrid retrieval. Paper 2.7 shows that plausible hidden-state query partitions need not improve retrieval: graph facets recover structure but can degrade evidence recall, making facet-to-query consolidation a key bottleneck. Paper 2.8 attacks memory-side compression: one mean gist can erase sparse native QK responses, yet compact native-key landmarks do not yield one universal controller. Paper 2.9 changes the *temporal extent of the semantic query* while holding downstream materialization fixed.

For tool names, API paths, URIs, headings, symbol tables, and other explicitly indexable sources, Paper 2.6 lexical/indexed retrieval is the default. Paper 2.9 targets natural-language composition, implicit relations, ambiguous phrases, and streaming generation where the complete query is not yet available.

2 Research questions

1. Does causal look-behind improve semantic retrieval over a current-token/current-state query at equal chunk budget?
2. How much oracle look-ahead gain can be recovered simply by delaying commitment until future tokens become history?
3. During prompt prefill, can a small non-causal routing sidecar exploit already-known future tokens without altering causal backbone states?

4. Is mean pooling enough, or is token-gist late interaction needed?
5. How does temporal query extent interact with one versus multiple chunk gists/landmarks?
6. Can routing run less frequently than token generation and be reused across layers?

3 Temporal routing

At layer ℓ and token t , let $H_{a:b}^\ell = \{h_a^\ell, \dots, h_b^\ell\}$. The baseline is $r_{t,\ell} = h_t^\ell$. Causal look-behind uses

$$r_{t,\ell}^B = R_B(H_{t-B+1:t}^\ell),$$

and oracle/prefill look-around uses

$$r_{t,\ell}^{B,F} = R_{B,F}(H_{t-B+1:t+F}^\ell).$$

Use $B \in \{1, 2, 4, 8, 16\}$ and $F \in \{0, 1, 2, 4, 8\}$. Whole-query routing is an upper/reference condition rather than the default streaming policy.

3.1 Query aggregation

Mean pooling is the minimal baseline,

$$\bar{q} = |W|^{-1} \sum_{i \in W} q_i.$$

Weighted/attention pooling uses $\sum_i a_i q_i$. Late interaction retains the query set and scores chunk c with representatives $g_{c,j}$:

$$S_{\text{LI}}(c) = \sum_{i \in W} w_i \max_j \cos(q_i, g_{c,j}).$$

Facet routing reduces the span to a small set $Z = \{z_1, \dots, z_f\}$ and applies a similar score. Paper 2.7 raw graph communities are a baseline, not an assumed winner; facets require consolidation and weighting.

3.2 Memory-side representations

Cross query methods with $m \in \{1, 2, 4, 8\}$ representatives per chunk. Begin with the inherited mean gist, then add local/prototype gists and compatible Paper 2.8 native-key landmarks. Do not confound temporal-query gains with a larger memory index.

4 Delayed commitment

A streaming router need not materialize memory at the first ambiguous token. Let $p_t(c)$ be the candidate distribution and H_t its entropy. Retain or prefetch candidates while uncertainty is high; commit when a validation-fixed entropy/margin threshold is crossed, a phrase boundary occurs, or maximum delay D is reached. Compare

$$\text{immediate}(B) \quad \text{vs} \quad \text{delayed}(B, D) \quad \text{vs} \quad \text{oracle/prefill}(B, F).$$

Delayed commitment is a causal approximation to look-ahead because future tokens eventually become observed history.

5 Prefill and predictive look-ahead

During prompt prefill all query tokens are known. Test a tiny local bidirectional routing sidecar over token embeddings or frozen states; it must not write into causal backbone states. For generation, predictive look-ahead is conditional on oracle headroom. Candidate methods are expected next-token embedding, a lightweight future-routing probe, and direct prediction of a coarse PRA destination. Every predictive method must beat an equal-latency delayed-routing baseline.

6 Slower routing clock

Test routing every $s \in \{1, 2, 4, 8\}$ tokens and at sparse layer checkpoints. Reuse the previous decision until confidence drops, selected identities change materially, or representation drift crosses a threshold. Report router calls/token, candidates scored/token, latency, and selected-memory churn.

7 Experimental design

Use inherited identity-disjoint HotpotQA, QASPER, 2WikiMultiHopQA, and MuSiQue cohorts where frozen features permit matched comparisons. Add controlled temporal-ambiguity fixtures: shared prefixes, compositional phrases, entity/relation ambiguity, order reversal, and distractor-heavy memories. Tool-like fixtures are mechanism controls only.

Required baselines are current-token semantic routing, global/whole-query semantic routing, Paper 2.6 exact/BM25/approximate/best hybrid where applicable, fixed windows, Paper 2.7 syntax/graph facets, and inherited one-mean-gist routing. Hold the final unique-chunk materialization budget fixed.

Primary metrics: evidence recall/precision, complete-evidence recovery, MRR, candidate entropy, candidate-set size, routing delay, selected-set churn, router calls, candidates scored, routing latency, and physical K/V materialized. Bootstrap identities rather than token windows.

8 Predeclared gates

- G0** Reproduce inherited matched baselines from Papers 2.6–2.8.
- G1** A causal look-behind condition beats $B = 1$ with paired 95% CI above zero on one natural dataset and no > 0.02 regression on the other primary dataset.
- G2** Delayed commitment recovers at least 70% of oracle look-ahead gain with median delay at most four tokens. If oracle look-ahead is negligible, predictive look-ahead is closed.
- G3** Prefill non-causal routing improves retrieval while disabled-PRA backbone logits/states remain unchanged.
- G4** Late interaction or multi-gist routing must justify its added routing cost relative to mean pooling.
- G5** Predictive future probes run only if substantial unrecovered oracle headroom remains and must beat equal-latency delayed commitment.
- G6** Native-K/V consumption/generation runs only after retrieval improvement survives matched budgets.

9 Falsification

A useful negative result is that h_t already retains enough history that explicit look-behind adds little. Another is that delayed commitment recovers nearly all oracle future gain, making predictive look-ahead unnecessary. If late interaction helps only with multiple memory representatives, memory-side compression rather than temporal query compression is the bottleneck. Explicitly structured sources are expected to remain better served by Paper 2.6 indexing.

10 Conclusion

Autoregressive generation is token sequential; semantic memory discovery need not be. PRA can preserve a causal frozen backbone while routing from several observed tokens, waiting for uncertainty to fall, or during prefill using a small non-causal sidecar. The decisive experiment is whether span-level routing improves retrieval at matched memory budget and whether causal delay recovers most apparent look-ahead benefit.